Trees 2 (Binary Lifting & Euler Tour)

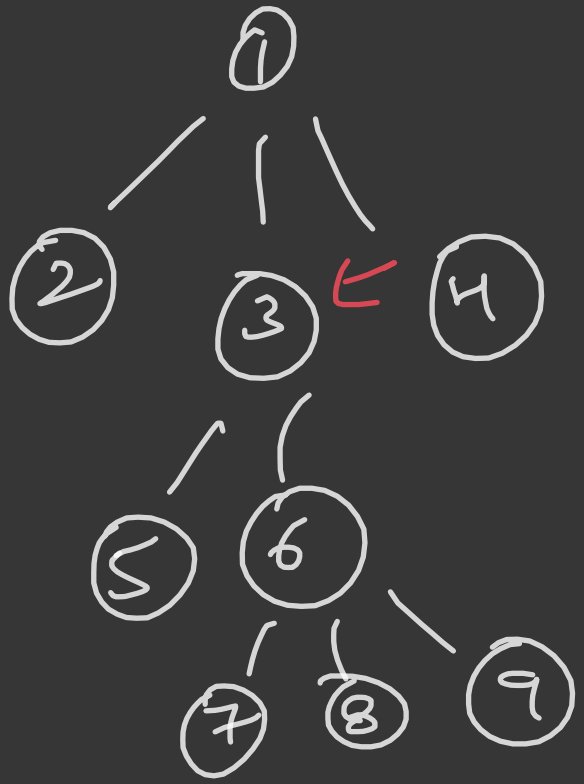
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Expert at codeforces (1817)
5 star at codechef (2040)

Binary Lifting & Euler Tour

- Print Kth parent of a node (n <= 1e5)
- Print Kth parent of a node (query based question q <= 1e5 & n <= 1e5)
- Binary lifting Intuition
- Print Kth parent of a node (query based question q <= 1e5 & n <= 1e5) using binary lifting
- LCA (Lowest Common Ancestor) using Binary Lifting
- Euler Tour (Tree flattening) - 2 types
- LCA using Euler Tour

Q. Print the kth parent of a node.

n → 1 e 5



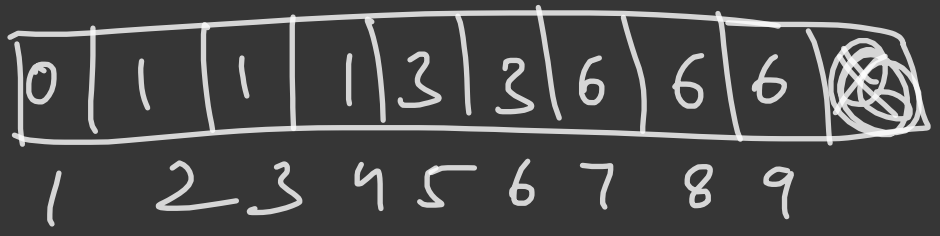
k = 2

node = 7

output → 3 ✓

node = 3

parent →



node = 7

```
while(k--){
    node = par[node]
}
return node;
```

Tc → O(n) + O(k) ≈ O(n)

Q 9 queries given

9 times [node K]

$q \rightarrow 1e^5$

$K \rightarrow n \rightarrow 1e^5$

$Tc \rightarrow O(n) + O(q \times K) \rightarrow TLE$

$1e^5 \quad 1e^5 \rightarrow 1e^{10}$ X

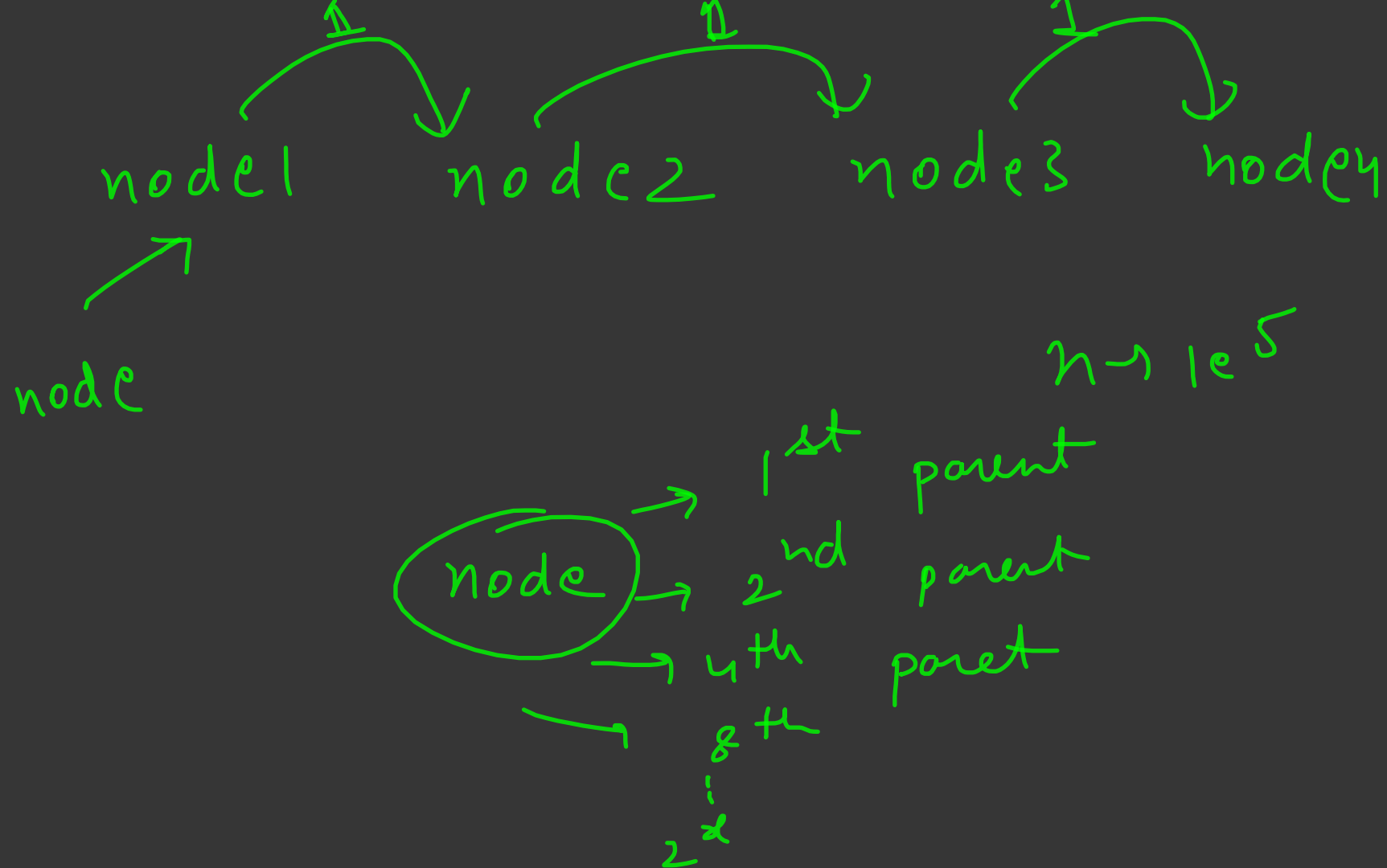
Intuition

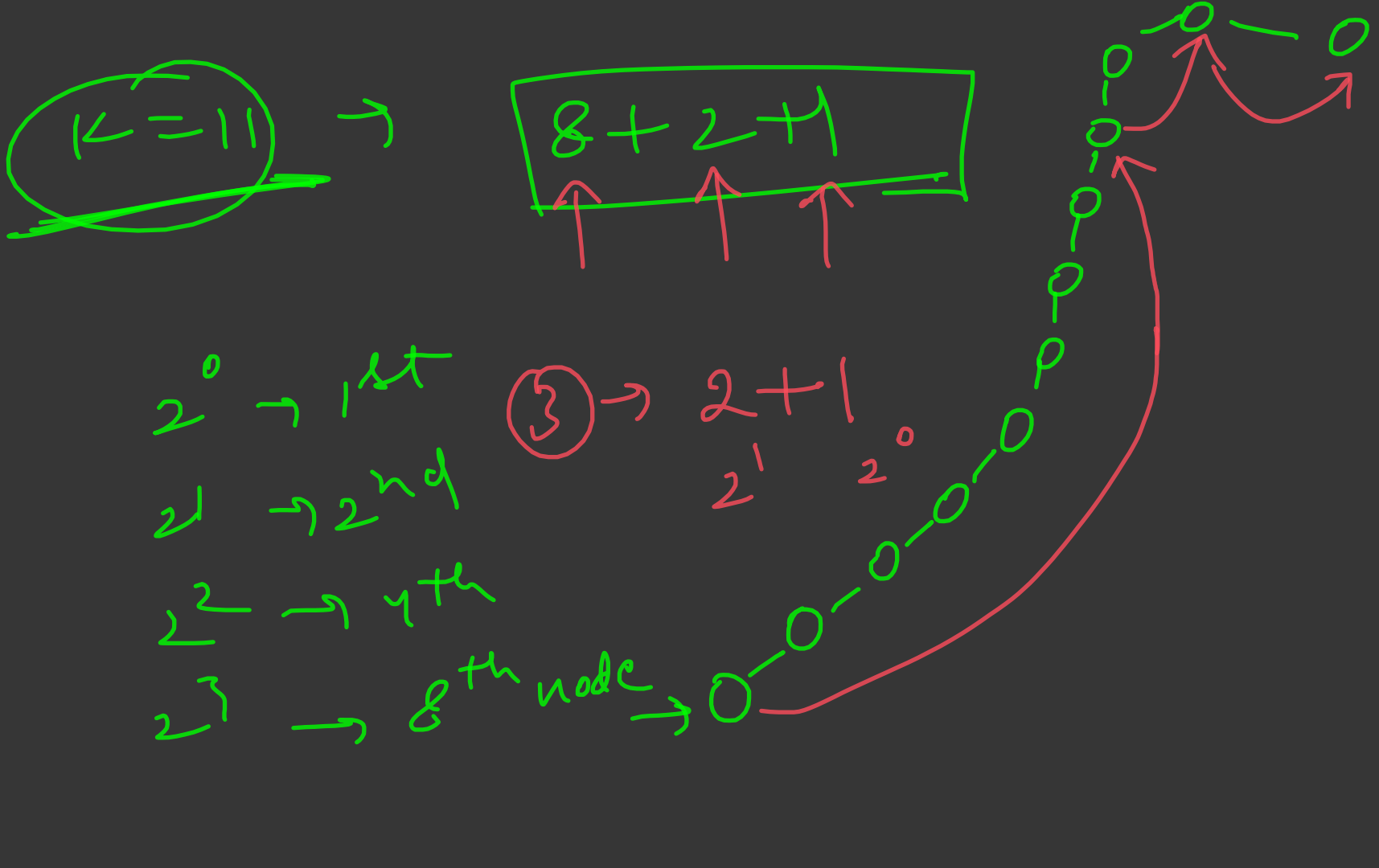
(K) integer

↳ this can be written as sum of power of 2.

Eg: $K = 11 \rightarrow 1 + 2 + 8 \rightarrow 2^0 + 2^1 + 2^3$

→ 1011

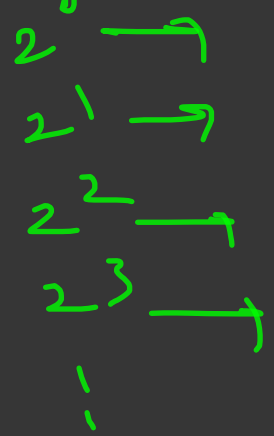

node1
node2
node3
node4
node
n → 1e 5
node
1st parent
2nd parent
4th poret
8th
2^d

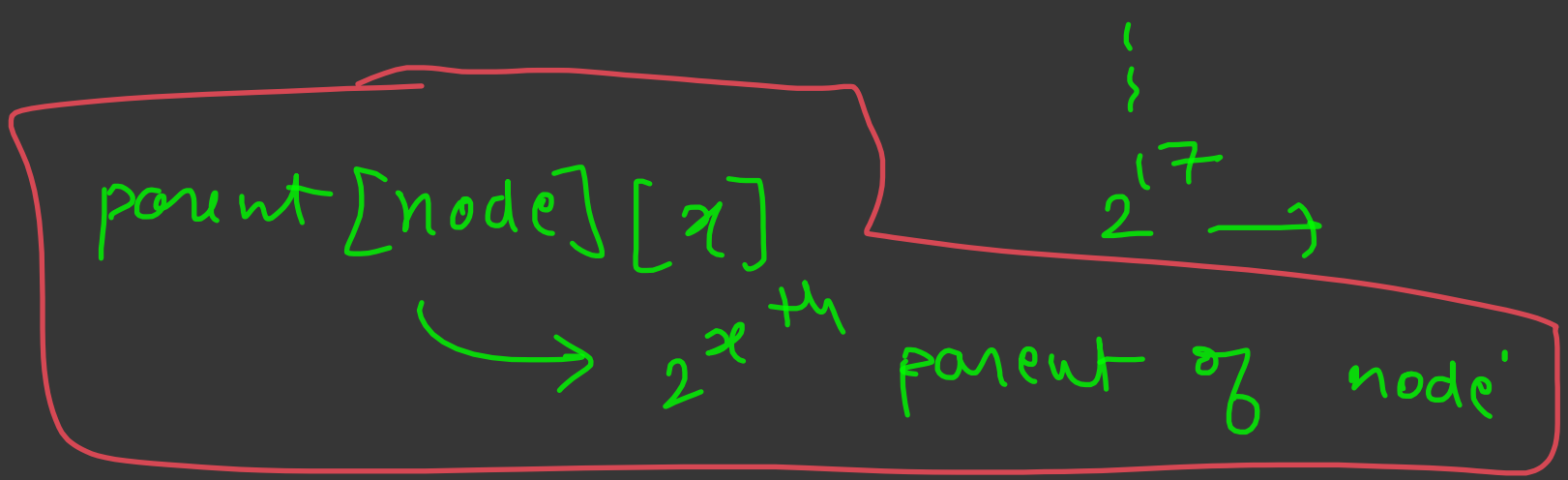

k=11
8+2+1
$2^0 \rightarrow 1^{st}$
$2^1 \rightarrow 2^{nd}$
$2^2 \rightarrow 4^{th}$
$2^3 \rightarrow 8^{th}$ node
3 → 2+1
2^1 2^0

$O(K) \longrightarrow O(\log k)$

Q how many parents per node?

$\log n$



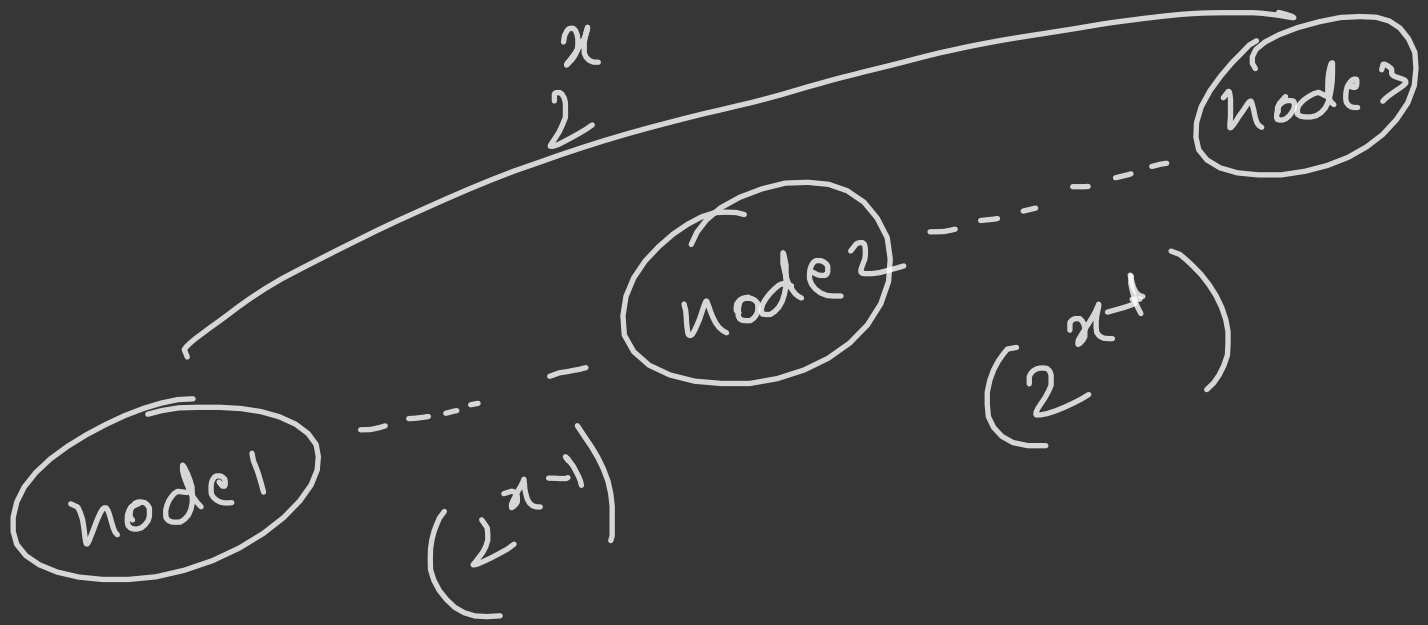


$x \rightarrow 17$

$node \rightarrow 10^5$

$10^5 \times 17$

$\rightarrow$ nlogn



parent [node1] [x-1] = node2

parent (node2] [x-1] = node3

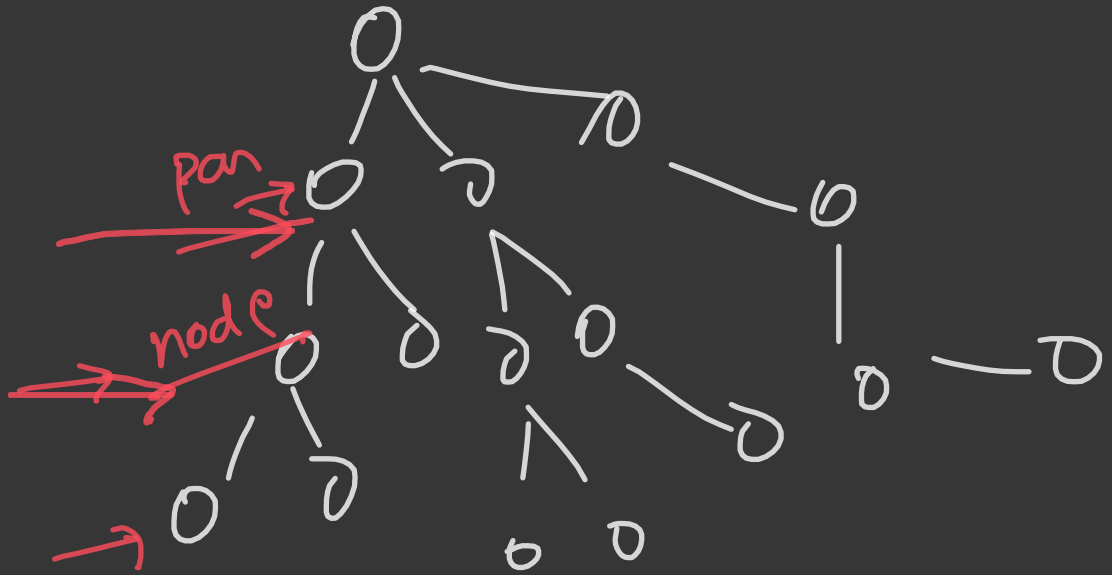
parent [node 1][x] = node 3

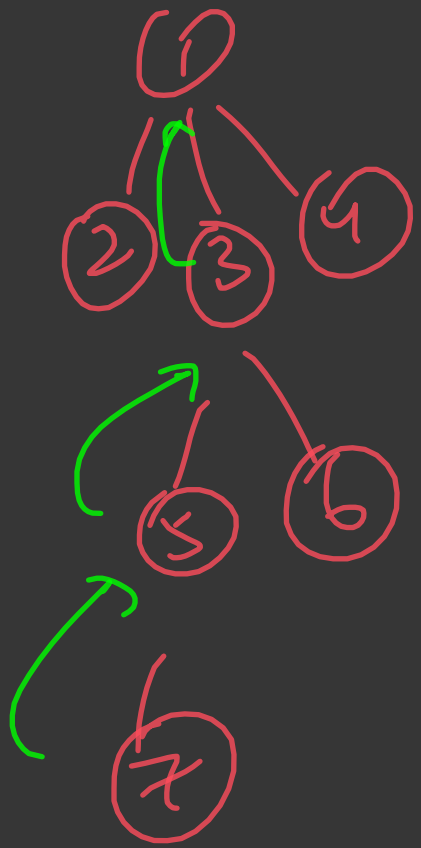
parent [node1][x] = parent [node2][x-1]

parent [node][x] = parent[parent(node][x-1]][x-1]

parent [node][1]

parent [node][0] → ✓ parent [node][2]


par
node



	0	1	2	3	4	— 11
1	0	0	0	0	0	0
2	1	0	0	0	0	0
3	1	0	0	0	0	0
4	1	0	0	0	0	0
5	3	1	0	0	0	0
6	3	1	0	0	0	0
7	5	3	0	0	0	0

dp

Q Now find k^{th} parent?

```
int cnt = 0;
while (k) {
    if (k & 1) {
        node = dp[node][cnt];
    }
    cnt++;
    k = k >> 1;
}
```

cnt = 2

k = 0

node → 1

```
}
return node;
```

 → 1 1 0 1

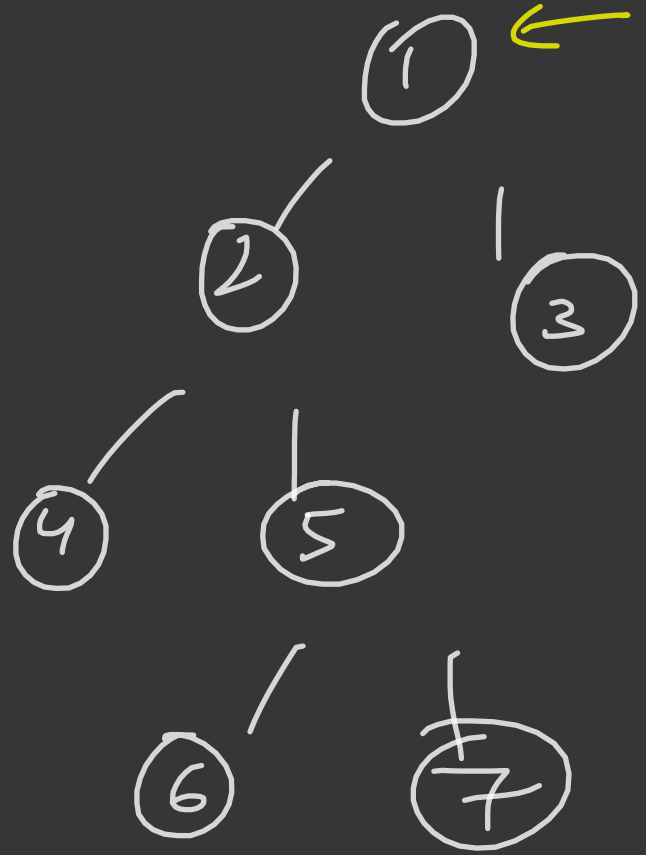
TC → $O(N \log N) + O(q \times \log K)$

dfs → $O(n \log n)$

LCA

4 → 2 → 1

5 → 2 → 1



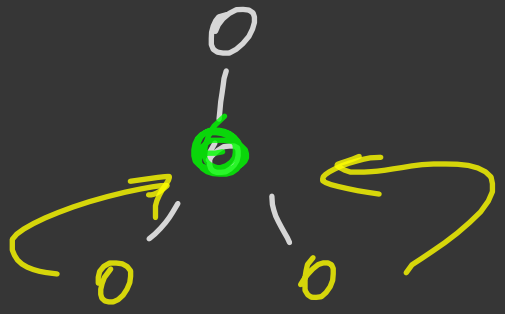
$lca(4,7) = 2$

a b

step 1 → bring them to the same level using binary lifting

step 2 → Move a & b to every unequal node (16 to 0)

$i = 0$



return dp[a][0]

or

dp[b][0]